



Review article

Prevalence and hazardous impact of pharmaceutical and personal care products and antibiotics in environment: A review on emerging contaminants

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ABSTRACT

Antibiotic resistance is a global health emergency linked to unrestrained use of pharmaceutical and personal care products (PPCPs) as prophylactic agent and therapeutic purposes across various industries. Occurrence of pharmaceuticals are identified in ground water, surface water, soils, and wastewater treatment plants (WWTPs) in ng/L to µg/L concentration range. The prevalence of organic compounds including antimicrobial agents, hormones, antibiotics, preservatives, disinfectants, synthetic musks etc. in environment have posed serious health concerns. The aim of this review is to elucidate the major sources accountable for emergence of antibiotic resistance. For this purpose, variety of introductory sources and fate of PPCPs in aquatic environment including human and veterinary wastes, aquaculture and agriculture related wastes, and other anthropogenic activities have been discussed. Furthermore, genetic and enzymatic factors responsible for transfer and appearance of antibiotic resistance genes are presented. Ecotoxicity of PPCPs has been studied in environment in order to present risk imposed to human and ecological health. As per published literature reports, the removal of antibiotics and related traces being difficult, couples the possibility of emergence of antibiotic resistance and hence sustainability in global water resources. Therefore, research on environmental behavior and control strategies should be conducted along with assessing their chronic toxicity to identify potential human and ecological risks.

1. Introduction

In the past few decades, ruthless use of antimicrobial agents and other organic compounds has been accelerated not only to treat bacterial infections but also in cancer treatment, heart surgery, and organ transplant (Hutchings et al., 2019). These pharmaceutical compounds work by utilizing either cytotoxic or cytostatic mechanism towards microorganisms, enabling natural immune system to exterminate them. Mechanism of antibiotics includes inhibition of cell wall synthesis, deoxyribonucleic acid (DNA) synthesis, ribonucleic acid (RNA) synthesis, protein synthesis, and membrane disorganization (Zaman et al.,

2017). The ability of antibiotics of being effective against broad categories of microbial strains makes them expedient to be employed worldwide. However, excessive utilization of antibiotics causes numerous human health and environmental issues. In a similar way, due to the presence of hazardous organic compounds in personal care products (PCPs), their usage has attracted attention of environmentalists to assess possible threats imposed on humans and ecology and are placed under the category of pharmaceutical and personal care products (PPCPs).

Small-scale treatment plants as well as waste-water treatment plants (WWTPs) are found unable to remove such pollutants efficiently and

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hence lead to dissemination of antibiotic resistance in aquatic system to a lethal level (Osińska et al., 2020). Persistence of PPCPs in water bodies and soil causes major environment threats as it leads to emergence of multi drug resistance (MDR) microbes, antibiotic resistant bacteria (ARB) and antibiotic resistant genes (ARGs). MDR is defined as the potential of microbial strains to resist the antibiotics despite being sensitive to it earlier (Chaturvedi et al., 2021 a & b). Bacterial strains have shown resistivity against various antibiotics such as carbapenems and cephalosporin resistance in *Klebsiella pneumoniae*; fluoroquinolones and cephalosporin resistance in *E. coli*; fluoroquinolones resistance in non-typhoidal *Salmonella*; methicillin resistance in *Staphylococcus aureus*; rifampicin resistance in *Mycobacterium tuberculosis* (MTB), and fluoroquinolones resistance in *Shigella* (Tanwar et al., 2014). According to a review antimicrobial resistance can cause 10 million deaths per year by 2050 (de Kraker et al., 2016).

Various factors are found responsible for emergence of MDR. The dogma conceived is the extensive and unregulated use of pharmaceutical substances in aquaculture, agriculture, veterinary, and human application (Shen et al., 2018). Metals like arsenic and cadmium has been identified as influential factors for causing antibiotic resistance in paddy soils (Zhao et al., 2020). Moreover, pharmaceuticals including erythromycin, sulfamethoxazole, trimethoprim, ibuprofen, diclofenac, carbamazepine, estriol, many others have been found to cause antibiotic resistance (Cizmas et al., 2015).

Resistance of pharmaceuticals is associated with morbidity and mortality around the world. Most of the bacterial strains present in aquatic system possess such ARGs, that can be incorporated in mobile platforms such as integron, transposon, and plasmid, thus can be easily transmitted to mutate naturally occurring bacterial consortium (Gheraout and Elboughdiri, 2020). For instance, efflux pump, horizontal gene transfer (HGT), plasmids, mutation, enzymatic modification of target, and extended-spectrum beta-lactamases (ESBL) producing strains have been recognized as the potent tool for conferring antibiotic resistance to microbes.

Anthropogenic activity has led to increase concentration of PPCPs that have been detected in various aquatic environments. The release of hazardous substances into landfill leachate due to disposal of municipal solid waste (MSW) and pharmaceuticals reaches to the nearby aquatic system and causes contamination to nearby sites. Numerous organic and inorganic compounds such as toxic metals, methane, ammonia, and carbon have been identified in landfill leachate (Peng et al., 2014). Varieties of PPCPs have also been detected in lake, river, sewage effluents, hospital and industrial effluents. The long-time exposure and persistence of PPCPs even at low concentration (low to high µg/L) can cause serious health threats. Exposure of PPCPs can cause emergence of ARGs, disruption of endocrine system, and other health impacts (Zhang et al., 2018b). Thus, the identification of occurrence, fate, transformation, transfer and subsequent management of PPCPs in environment is an important issue to tackle.

PPCPs and endocrine disruptors including 17β-estradiol, 4-Nonylphenol, Testosterone, Simazine, Atrazine, Diethylstilbestrol, 17α-ethinylestradiol, Bisphenol A, and Carbamazepine in environment have been explored and are categorised into hormones, conjugates and metabolites, antidepressants, anticonvulsants, non-steroidal anti-inflammatory drugs, lipid regulators, and antihypertensives (Cizmas et al., 2015). Due to the unavoidable and accelerated use PPCPs have been monitored in river water, drinking water, soils and sediments, and other water resources (Xu et al., 2019). The investigation of PPCPs and their exposure to humans and animals has linked them to numerous reproductive, mutagenic, and carcinogenic toxicity (Catanese et al., 2015; Kiyama and Wada-Kiyama, 2015). PPCPs can interfere with hormonal system and also found to cause oxidative stress, histological and cytological changes, and inhibition of hormones and enzyme in vertebrates (Cizmas et al., 2015).

Many researchers are continuously working to assess chronic effects linked to PPCPs on ecology and human health due to their occurrence

and persistence in aquatic and soil environment. A data representing number of researches published year and country-wise has been depicted in Fig. 1(a) and (b) and was constructed using Scopus database. Graphs reveal a comparative study summarizing work performed on occurrence of pharmaceuticals and personal care products from year 2009–2019 and top 10 countries, demonstrating need of research to elevate awareness regarding environmental pollutants in high populated countries like India and China.

This paper conceptualizes and describes major sources for occurrence and fate of PPCPs responsible for antibiotic resistance in environmental bacteria. Development of ARGs and ARB is a main concern of biotechnologists and thus various possible influential impacts have been investigated. This review summarizes various factors responsible for resisting antibiotics and related products in microbiota. In this view, many discrete genetical, enzymatical, and environmental factors have been discussed along with case studies to validate the impact of PPCPs on resistance emergence. Risk associated with persistence and prevalence of organic pollutants in environment and their effect on ecology and human health have been presented.

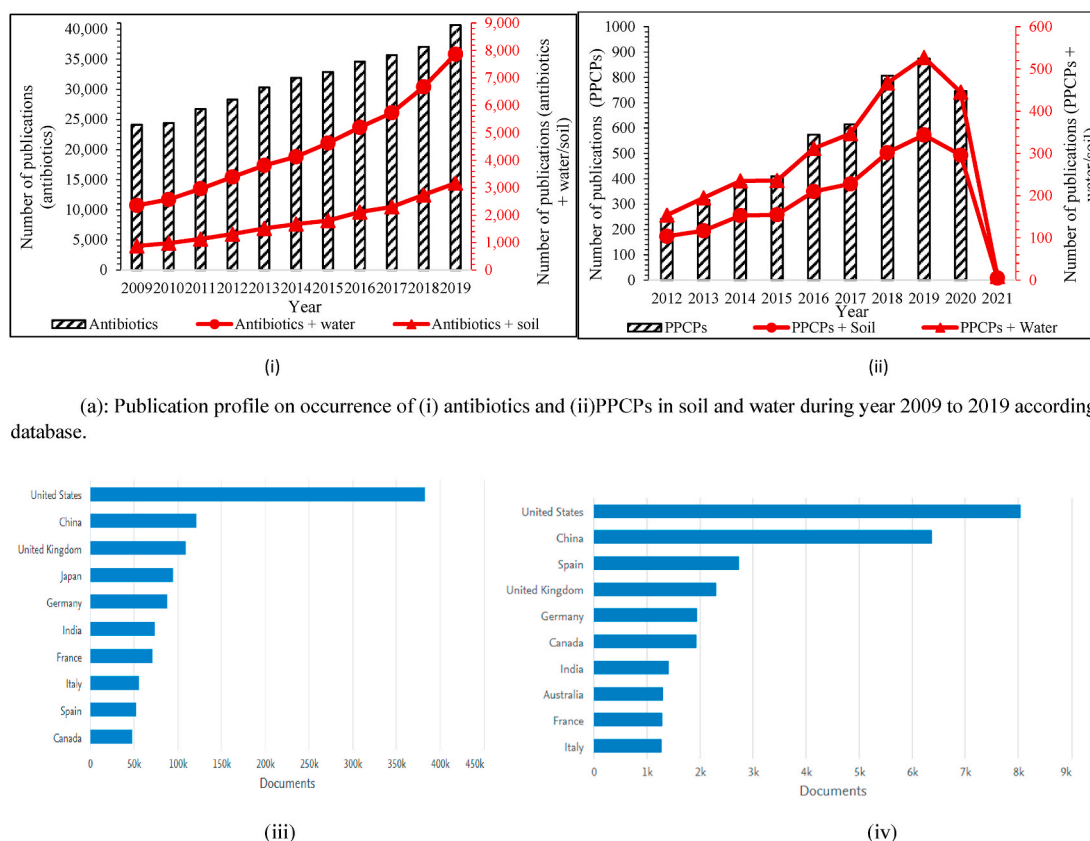
2. Accumulation, fate, and source of emerging contaminants in environments

Addition of micro-pollutants to ecosystem is a continuous process and it usually depends upon active concentration present among the population, manufacturing etc. Generally, rate of organic pollutant dispersion can be enhanced by anthropogenic impacts. Use of synthetic organic compounds in health related prospective and personal care products have led all time serious deleterious concentration in environment (Liu and Wong, 2013a). Implication of PPCPs in human health care, agriculture, aquaculture, and veterinary and their discharge in same or low altered form causes the emergence of highly resistant pathogens. Occurrence of PPCPs in environmental matrices and their respective concentrations have been reported in Tables 1 and 2.

An overview demonstrating potential sources of PPCPs and probable reason to be found in ground and waste water treatment plants ultimately causing emergence to antibiotic resistance bacteria, have been depicted in Fig. 2.

2.1. In human and veterinary wastes

To minimize threat and burden of fatal bacterial diseases, pharmaceuticals are frequently used in human and veterinary treatments. Approximately 100,000 to 200,000 tons antibiotics are consumed per year that has recorded an increase of 36% leading to global use of 73.6×10^9 from 54.1×10^9 standard unit (Dinh et al., 2017). Numerous antibiotics including doxycycline, fluoroquinolones, penicillins, lincomycin, trimethoprim, bacitracin, colistin, chloramphenicol, erythromycin and cephalosporin are the majorly utilized (Checcucci et al., 2020). Also, to meet the huge demand of increasing population pharmaceutical industries are accelerating drug production. Some of these get consumed and part of it excreted out unaltered in urine and feces by animals and human. Sewage treatments plants (STPs) have been majorly identified as reservoirs of antibiotic traces and PPCPs with concentration up to µg/L. In most cases concentration from not-detected (ND) to 7910 ng/L (influent) and not-detected to 9460 ng/L (effluent) has been quantified in STPs, China (Liu and Wong, 2013a). Till date several antibiotics (macrolides, fluoroquinolones, sulfonamides, azole antifungal drugs, cephalosporins, erythromycin) (Peng et al., 2012), their metabolic products (erythromycin-H₂O, roxithromycin (metabolites of erythromycin) (Liu and Wong, 2013a), hormones (estrogen, progesterone, androgens, glucocorticoids) (Zhang et al., 2018b), other pharmaceuticals (acetaminophen, caffeine, naproxen, ibuprofen) were found in STP sewage samples (Wu et al., 2014). Adsorption and biodegradation are the two main processes contributing to fates of PPCPs in STPs. Similarly, investigation of waste water revealed PPCPs emanating from hospital



(a): Publication profile on occurrence of (i) antibiotics and (ii) PPCPs in soil and water during year 2009 to 2019 according to Scopus database.

(b): Total researches published on (iii) antibiotics and (iv) PPCPs in various countries according to Scopus database search.

Fig. 1. (a): Publication profile on occurrence of (i) antibiotics and (ii) PPCPs in soil and water during year 2009–2019 according to Scopus database. (b): Total researches published on (iii) antibiotics and (iv) PPCPs in various countries according to Scopus database search.

effluents. A study performed in South Africa found bezafibrate, ofloxacin, and chloramphenicol below limit of quantification (LOQ). Whereas, dominance of ibuprofen (0.3–63 µg/L) and acetaminophen (21–119 µg/L) was identified (Kanama et al., 2018). Another study, also documented high load of PPCPs including gabapentin (10–17 µg/L), metformin (30–720 µg/L), ibuprofen, and caffeine (86 µg/L) in hospital effluent laden wastewater treatment plant (Oliveira et al., 2015). Study performed in countries of Latin America (Mexico) and South America (Brazil) demonstrated the presence of ARGs and contaminants including imipenem, amoxicillin, estrone, ampicillin, bisphenol-A, di(2-ethestradiylhexyl) phthalate, estradiol, triclosan, 4-nonylphenol, estrone, ethinylestradiol, and norfloxacin in potable water, drinking water, sewage water, and hospital effluents (Reichert et al., 2019).

In livestock, tetracycline, quinolones, penicillin, amoxicillin, gentamicin, tylosin, and novobiocin are most frequently used. Diarrhoea, bovine pneumonia, and shipping fever are the most treated diseases by AMA in meat producing animals. Antibiotics including fenbendazole (3.85–241 µg/L), P-hydroxyfenbendazole (0.693–92.7 µg/L) in livestock farms (Sim et al., 2013), sulfamethazine (0.018 µg/L) in pig farms, sulfamerazine, sulfathiazole, lincomycin, erythromycin, tiamulin, sulfadimethoxine in swine and cattle beef facilities (Bartelt-Hunt et al., 2011), oxytetracycline (2 ng/L–68 µg/L) in beef cattle, dairy cattle, chicken and swine farms, sulfamethazine (18500–19200 ng/L) in pig farms, monensin (180–2350 ng/L), ractopamine (50 ng/L) in swine and cattle farms (Bartelt-Hunt et al., 2011), tetracycline, sulfonamides in swine manures, sulphasalazine (202–321 ng/L), oxytetracycline (109–423 ng/L) in dairy farms, anthelmintics (0.32–39.43 ng/L) in pig farms, oxytetracycline (2 ng/L–68 µg/L), sulfamethoxazole (20–174 ng/L), trimethoprim (7–44 ng/L) in chicken farms have been identified (Kaczala and Blum, 2016).

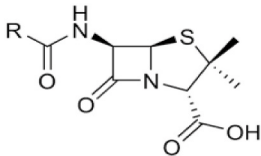
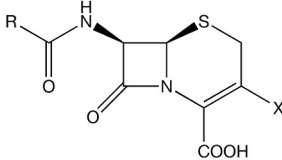
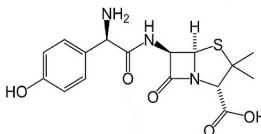
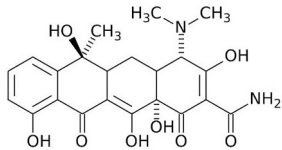
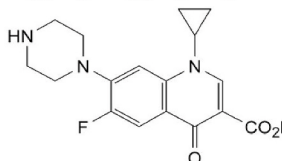
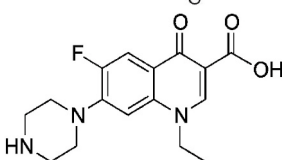
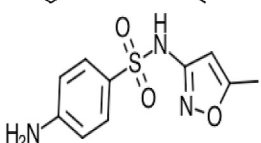
Moreover, antimicrobial peptide colistine (polymyxin E) is widely utilized for veterinary and human consumption. Colistine is extensively used in food animals specifically in swine to control Enterobacteriaceae infections. However, over usage has caused plasmid mediated colistine resistance by *mcr-I* gene (Rhouma et al., 2016). For instance, active concentration of colistine has been identified in animal manures (Quaik et al., 2020). In addition, presence of florfenicol, cyromazine, tetracycline, sulfonamides has been found in manure and manure laden soil (Wei et al., 2016).

2.2. In agriculture and aquaculture related products

Use of antimicrobial agents (AMAs) and PPCPs are not only constrained in prevention of diseases but also employed in animal husbandries and agriculture as growth promoters and to increase feed conversion efficiency (Cheng et al., 2016). The type of antibiotic utilized is influenced by several factors including geographical terrain, crop to be produced, food animal species, production pattern, farming purpose (industrial, commercial or domestic), nature of farming (extensive or intensive), population size and demand, and farmer's choice in particular (Manyi-Loh et al., 2018). Although utilization of antibiotics is acknowledged as deleterious, its employment in crop production also causes serious risks. Glyphosate, patented as antibiotic, is commonly used as weed killer in agriculture. Its adverse effect has been studied on shikimate pathway, a mechanism to produce essential amino acids in microbes and plants. Moreover, usage of pesticide in crop improvement is also associated with antibiotic resistance emergence (Rother, 2008). In addition, oxytetracycline and streptomycin are two mostly used antibiotics in apple cultivation, vegetable production, and floriculture production. The application of treated water containing PPCPs for

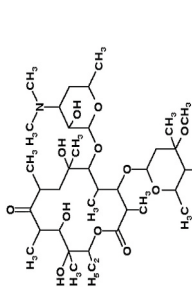
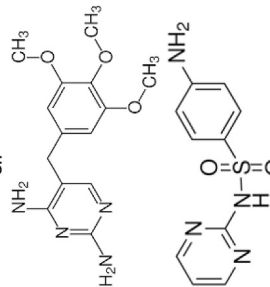
Table 1

Physiochemical characteristic, related class, target disease, presence in environmental matrices, their concentration and side-effects of antibiotics on human health.

Order	Antibiotic	Chemical formula	Structure of antibiotic	Clinical Use	Class	Presence in environment/ concentration	Side-effects on human health	References
1	Penicillin G	$C_{16}H_{18}N_2O_4S$		Pneumonia, infections, meningitis, anthrax, gonorrhoea, and syphilis	β -Lactam	Liquid manure, Soil/ 48 ngL^{-1}	Diarrhoea, type-I hypersensitivity, type-III hypersensitivity (serum sickness), nausea, rash, neurotoxicity, fever, vomiting, angioedema, seizures	(Christian et al., 2003; Hitchings et al., 2018)
2	Cephalosporin	$C_{15}H_{21}N_3O_7S$		Pneumonia, bone and joint infections, urinary tract infections, gynaecological infections, and septicaemia	β -Lactam	Surface water, Sewage water/ 0.30 ngL^{-1} to 0.03 mgL^{-1}	Perioperative anaphylaxis, maculopapular or morbilliform rash, positive coomb's test, urticaria, serum sickness-type reaction	(Khan et al., 2019; Ribeiro et al., 2018)
3	Amoxicillin	$C_{16}H_{19}N_3O_5S$		Genitourinary tract infections, tonsillitis, lower respiratory tract infection, pharyngitis, ear, nose, and throat infection	β -Lactam	Sewage treatment plants effluents, Surface water, Hospital wastes, WWTPs/ $1.8\text{--}1670 \text{ ngL}^{-1}$	Nausea, vomiting, diarrhoea, type-I, II, III, IV hypersensitivity reactions	(Elizalde-Velázquez et al., 2016; Gillies et al., 2015)
4	Tetracycline	$C_{22}H_{24}N_2O_8$		Respiratory tract infection, urogenital, and gastrointestinal tract infection	Tetracycline	Liquid manure/ 20 and 40 mgL^{-1} , Ground water/ $0.5 \text{ }\mu\text{gL}^{-1}$	Loss of appetite, black hairy tongue, nausea, diarrhoea, sore throat, dizziness, headache	(Grossman, 2016; Mackie et al., 2006)
5	Ciprofloxacin	$C_{17}H_{18}FN_3O_3$		Neonatal sepsis, urinary tract infection, cystic fibrosis, typhoid, and diarrhoea	Fluoroquinolones	Tap water/ 1.0 to 679.7 ngL^{-1} and 2.0 to 37 ngL^{-1} , River water/ 3 ngL^{-1}	Nausea, vomiting, tiredness, pale skin, abnormal liver, headache	(Drossou-Agakidou et al., 2004)
6	Norfloxacin	$C_{16}H_{18}FN_3O_3$		Genitourinary tract infection	Fluoroquinolones	Tap water/ 1.0 to 679.7 ngL^{-1} and 2.0 to 37 ngL^{-1} , River water/ 3 ngL^{-1}	Muscle and joint pain, nausea, headache, rectal pain, dizziness, diarrhoea	(Qandil et al., 2014)
7	Sulfamethoxazole	$C_{10}H_{11}N_3O_3S$		Respiratory tract and urinary tract infection, shigellosis	Sulfonamides	Ground water/ 0.01 to 3460.57 ngL^{-1} , River water/ 326 ngL^{-1} , Plant leaves/ $0.9\text{--}2.7 \text{ mgkg}^{-1}$, Sewage water/ $119\text{--}544 \text{ ngL}^{-1}$	Insomnia, fatigue, nervousness, apathy, rash, pruritus, neuropathy, headache	(Masters et al., 2003; Raz et al., 2002)
8	Erythromycin	$C_{37}H_{67}NO_{13}$		Respiratory tract infection including bronchitis, pneumonia, pertussis, diphtheria	Macrolides	Tap water, River water/ $1.0\text{--}320.5 \text{ ngL}^{-1}$, Sewage water/ $20\text{--}300 \text{ ngL}^{-1}$	Nausea, vomiting, cramping, diarrhoea, upper abdominal pain, loss pf appetite	(Gothwal and Shashidhar, 2015)

(continued on next page)

Table 1 (continued)

Order	Antibiotic	Chemical formula	Structure of antibiotic	Clinical Use	Class	Presence in environment/ concentration	Side-effects on human health	References
9	Trimethoprim	$C_{14}H_{18}N_4O_3$		Urinary tract infections (cystitis)	Sulfonamides	Ground water/3.3–10.5 ngL^{-1}	Vomiting, nausea, diarrhoea, headache, loss of appetite, changes in taste	Peng et al. (2014)
10	Sulphadiazine	$C_{10}H_{10}N_4O_2S$		Toxoplasma gondii encephalitis, malaria, urinary tract infections, ear infection and chancroid	Sulfonamides	Ground water/Non-detectable –208 ngL^{-1}	Vomiting, nausea, diarrhoea, headache, loss of appetite	López-Serna et al. (2013)

irrigation in arid and semi-arid areas pose unknown risk to consumers. The study on translocation and accumulation of PPCPs has revealed the presence in roots, leaves, and fruit. Various PPCPs such as fluoxetine, diazepam, triclosan, triclocarban in roots and carbamazepine, primidone, diuron, meprobamate, and dilantin, in leaves of lettuce, cucumber, pepper, and spinach were found (Wu et al., 2013). The uptake of four PPCPs and endocrine disrupting chemicals (EDCs) including 4-nonylphenol, diclofenac sodium, bisphenol A, and naproxen were found in collards and lettuce (Dodgen et al., 2013). Moreover, the ability of oxytocin hormone to enhance crop yield, and growth (bottle guard (*Lagenaria siceraria*), cucumber (*Cucumis sativus*), bitter guard (*Momordica charantia*), winter wheat (*Triticum sativum*) and bent (*Agrostis tenuis*), lettuce (*Lactuca sativa*), pumpkins (*Cucurbita maxima*)) has increased its application in vegetation. Hence, its absorption in human and animals has speculated adverse health effect such as harm on reproductive system and decreased life spans in cattle, and hormonal imbalance, cases of abortion, and haemorrhage in humans (Lawson, 2017).

To promote the growth and provide protection from bacterial attack, antibiotics are being exercised in aquaculture. Government guidelines and legislations, size of biomass intended, disease encountered, duration of treatment, and physical parameters (temperature, salinity, and photoperiod) are prime governing factors of antibiotic application (Watts et al., 2017). Penicillin, sulfonamide, tetracycline, phenicol, and quinolone are extensively used. In which phenicol, tetracycline, and quinolone are mostly applied in salmon farming. Excessive use of antibiotics in aquaculture hampers the aquatic environment as adverse effects on growth and development of algae has been identified due to the presence of macrolides, sulfonamides, and tetracycline (Gao et al., 2013). In liver, gill, and muscle of fish high content of steroid estrogens (11.3 ng/g) was found (Liu et al., 2011). Accumulation of musk fragrances was found higher in adipose tissues of Chinese sturgeon (*Acipenser sinensis*), confirming its high affinity towards lipids (Wan et al., 2007).

(Zhao et al., 2018) revealed the presence of ARGs in gut of aquatic shrimp using metagenomic analysis, which showed that the target modification and efflux pump were predominantly associated with emergence of resistance. Morphological and transcriptomic mutations in fish kidney (Bickley et al., 2017) and anti-ovulatory outcomes in vertebrates (aquatic) (Yokota et al., 2018) have been observed because of diclofenac (DIC). A study performed on aquatic sample demonstrated the presence of *tet* (C) and *sul* I ARGs (Yuan et al., 2019). Another research performed by Wang et al. (2015) demonstrated the hazardous effect of manure-amended soil for the growth of edible plants which shows the presence of ARGs and *int1* in harvested vegetables.

2.3. Occurrence in soil and sediments

Severe uses of PPCPs have been linked humans and animals to an array of dreadful mutagenic and carcinogenic substances. Various prescribed, non-prescribed drugs in human and animal medicine, other chemicals present in cosmetics like shampoos, sunscreen, perfumes, beauty creams, hair colours are considered under PPCPs. PPCPs contamination in environment is generally associated with agricultural practices, municipal wastes, discharge of used substances from industries, hospitals, livestock husbandries, houses, and leaching from disposed pharmaceuticals (Bartrons and Peñuelas, 2017). Activeness in biological systems and occurrence of PPCPs even in low concentration is associated with extreme effects in organisms. With the application of fertilizers, landfill, and irrigation, pharmaceuticals are introduced in soil. Antibiotics (norfloxacin, erythromycin, sulfadimethoxine, chloramphenicol), hormones (estradiol, estrone), anti-inflammatory drugs and analgesics (ibuprofen, acetylsalicylic acid), contrast media (diatrizoate), antimicrobial agents (triclosan), fragrances/synthetic musks [Galaxolide (HHCb), toxalide (AHTN)], preservatives (parabens), insect repellent [N,N-diethyl-m-toluamide (DEET)], sunscreen UV filters

Table 2

A summary of toxicity imposed by Pharmaceutical and personal care products in different countries, their concentration in environmental matrices, and respective class.

Order	PPCPs	Acronym	Chemical formula	Class	Presence in environmental matrix	Concentration	Country	References
1.	Ibuprofen/ Diclofenac	IPF/DCF	C ₁₃ H ₁₈ O ₂ / C ₁₄ H ₁₁ O ₂ Cl ₂ NO ₂	NSAID	Ground water/ ground water and sea water	0.16–200 µg L ⁻¹ / 0.184–225 µg L ⁻¹ , 9.7 µg L ⁻¹ , >3 µg L ⁻¹	Spain/ Spain, France, Saudi Arabia	(Ali et al., 2017; López-Serna et al., 2013; Vulliet and Cren-Olivé, 2011)
2.	Naproxen/ Indomethacin	NPX/IMC	C ₁₄ H ₁₄ O ₃ / C ₁₉ H ₁₆ ClNO ₄	NSAID	Ground water/ Ground water	67.0 ng L ⁻¹ , ND- 0.43 ng L ⁻¹ , 1.2 ng L ⁻¹ /3.5–11.7 ng L ⁻¹	China, Spain, France/ China	(López-Serna et al., 2013; Peng et al., 2014; Vulliet and Cren-Olivé, 2011)
3.	Caffeine/Nicotine	CAF/NIC	C ₈ H ₁₀ N ₄ O ₂ / C ₁₀ H ₁₄ N ₂	Stimulant	Sea water, River water/treated and untreated WWTPs	62–>3000 ng L ⁻¹ , 865 ng L ⁻¹ / 85.7–3919.3 ng L ⁻¹	Saudi Arabia, China, England, UK	(Ali et al., 2017; Baker and Kasprzyk-Hordern, 2013; Fu et al., 2019)
4.	Acetaminophen/ Paracetamol	ACE/ PCM	C ₈ H ₉ NO ₂ /C ₈ H ₉ NO ₂	Analgesic and antipyretic	Sea water, surface water, sediment/ Municipal and hospital WWTPs	<LOQ-2379 ng L ⁻¹ , 71.95 ng L ⁻¹ , 23.98 ng L ⁻¹ /20.6 ng L ⁻¹ , 9.3 µg L ⁻¹	Saudi Arabia, China, Greece	(Ali et al., 2017; Kosma et al., 2010)
5.	Metformin	MET	C ₄ H ₁₁ N ₅	Anti-diabetic (antihyperglycemic agent)	Sea water,	7– >3000 ng L ⁻¹	Saudi Arabia	Ali et al. (2017)
6.	Triclocarban/ Triclosan	TCC/TCS	C ₁₃ H ₉ Cl ₃ N ₂ O/ C ₁₂ H ₇ Cl ₃ O ₂	Antimicrobial	WWTP (influent)/ WWTP (influent)	515–8880 ng L ⁻¹ / 145–4890 ng L ⁻¹ ,	India	Balakrishna et al. (2017)
7.	Diltiazem	DTZ	C ₂₂ H ₂₆ N ₂ O ₄ S	non-dihydropyridine calcium channel blockers	Surface water, sediment	13.88 ng L ⁻¹ , 15.27 ng L ⁻¹	China	Zhang et al. (2018a,b)
8.	Carbamazepine	CBZ	C ₁₅ H ₁₂ N ₂ O	Anticonvulsant	Surface water, sediment	271.02 µg L ⁻¹ , 54.20 µg L ⁻¹ , 6.3 µg L ⁻¹	China, Germany	Zhang et al. (2018a,b)
9.	Morphine/ Dihydrocodeine	MOP/ DHC	C ₁₇ H ₁₉ NO ₃ / C ₁₈ H ₂₃ NO ₃	Opioid analgesics	River water, treated and untreated WWTPs	59.1–371.2 ng L ⁻¹ / 118.2–226.6 ng L ⁻¹	England, UK	Baker and Kasprzyk-Hordern (2013)
10.	17α- ethinylestradiol/17 β-estradiol	EE2/E2	C ₂₀ H ₂₄ O ₂ / C ₁₈ H ₂₄ O ₂	Estrogen (endocrine disrupting compounds)	Sediment and sludge/WWTPs	48.1 µg Kg ⁻¹ /3.9 ng L ⁻¹	Spain/ California, USA	(Ebele et al., 2017; Jjemba, 2019)
11.	Methylparaben/ Ethylparaben	MP/EP	C ₈ H ₈ O ₃	Preservative (endocrine disrupter)	River water, reservoirs	1192.39 mg L ⁻¹ , 8 mg L ⁻¹ , 27.50 mg L ⁻¹ /12.5 ng L ⁻¹	Brazil, China	(Galinario et al., 2015; Peng et al., 2014; Pompei et al., 2019)
12.	Clofibric acid/ Benzafibrate/ Gemfibrozil	CFA/ BZF/GFZ	C ₁₀ H ₁₁ ClO ₃ / C ₁₉ H ₂₀ ClNO ₄ / C ₁₅ H ₂₂ O ₃	Lipid regulator	Ground water	73.9 ng L ⁻¹ /ND- 7.57 ng L ⁻¹ , ND- 25.8 ng L ⁻¹	Spain	López-Serna et al. (2013)
13.	Clotrimazole/ Econazole/ Miconazole/ Tebuconazole/ Ketoconazole	CTZ/ ECZ/ MCZ/ TCZ/KCZ	C ₂₂ H ₁₇ ClN ₂ / C ₁₈ H ₁₅ Cl ₃ N ₂ O/ C ₁₈ H ₁₄ Cl ₄ N ₂ O/ C ₁₆ H ₂₂ ClN ₃ O/ C ₂₆ H ₂₈ Cl ₂ N ₄ O ₄	Azole anti-fungal and pesticides	Ground water, reservoirs	1.5 ng L ⁻¹ /LOQ/ 6.7 ng L ⁻¹ /0.8 ng L ⁻¹ /3.3 ng L ⁻¹	China	Peng et al. (2014)

ND: Not detected above limit of quantification; LOQ: Limit of quantification; NSAID: Non-steroidal anti-inflammatory.

(2-ethyl-hexyl-4-trimethoxycinnamate (EHMC), 4-methyl-benzilidene-camphor (4MBC) have been majorly identified as PPCPs and detected in soils (Liu and Wong, 2013b).

For instance, effect of ciprofloxacin in soil has been studied and was found to retard growth and caused morphological deformities in higher plants by altering photosynthetic pathways (Guruge et al., 2019). Several studies have revealed the persistence of pharmaceuticals in sediments. Pharmaceuticals including erythromycin, sulfadiazine, clarithromycin, sulfamethoxazole, ibuprofen, naproxen, ketoprofen, clofibric acid, indomethacin, gemfibrozil in Germany, triclocarban (822 ngg⁻¹), caffeine, nicotine, carbamazepine, lidocaine, ephedrine in North and South America, while presence of estrogens including estriol, estrone, β-estradiol/17-valerate, β-estradiol, diethylstilbestrol, and 17α-ethynylestradiol ranging from 0.98 to 51.6 ngg⁻¹ and antibiotic ofloxacin (1560 µgkg⁻¹) in China, and triclosan concentration of 32.1 ngg⁻¹, 46.9 ngg⁻¹, 85.3 ngg⁻¹ in Vellar, Tamiraparani, Kaveri river, respectively in India have been detected (Ebele et al., 2017). Another study performed by (Chen et al., 2013a) identified the presence of

diclofenac sodium, ibuprofen, naproxen, and clofibric acid in soil in California, United States. In North Carolina, U.S.A., study performed on municipal waste water irrigated soil demonstrated the presence of 33 PPCPs compound including atenolol, sulfamethazine, paraxanthine, cotinine, caffeine, salicylic acid, trimethoprim, salicylic acid, naproxen, diltiazem, estriol, triamterene, meprobamate, atenolol, testosterone, gemfibrozil, cholesterol, fluoxetine, progesterone, diphenhydramine, and many more at a concentration from 8.45 to 13.07 ngg⁻¹ (Harbordt, 2016). In biosolid-amended soil, azole biocides were identified having 440 days of half-lives, demonstrating high potential of persistence. In addition, miconazole, clotrimazole, and climbazole have also been detected in biosolid-amended soil (Chen et al., 2013b). Moreover, presence of antibiotic and PPCPs traces has been detected in microorganisms and fauna. According to a research, PPCPs exert adverse effect of soil organism by destroying their population hence affecting soil bacterial biomass. Also, PPCPs may potentially affect the metabolic activity of microorganisms subsequently altering their ability of nitrification, enzyme activities, and degradation properties. For example,

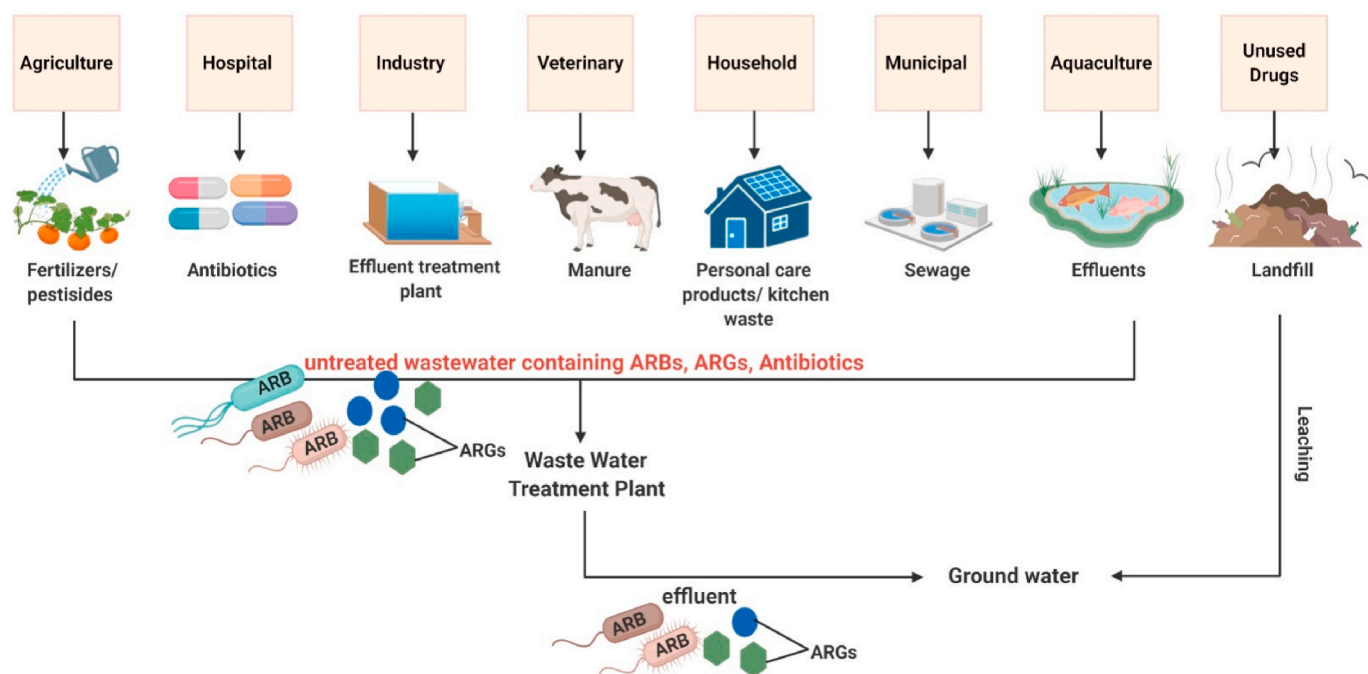


Fig. 2. representation of different source with their possible effluent produced conferring emergence of antibiotic resistant genes (ARG) and antibiotic resistant bacteria (ARB).

inhibition of respiration has been identified due to presence of sulfamethoxazole and ciprofloxacin (Qin et al., 2015).

2.4. Occurrence in aquatic system

PPCPs are ubiquitously found in various environmental matrices. In most cases PPCPs are found in excretion products (urine, faeces) and thus are most commonly found in WWTPs, which eventually entered in aquatic system.

Yang et al., 2013 demonstrated the presence of PPCPs in both untreated and treated water in Shijing river, China. In this study ciprofloxacin, acetaminophen, and caffeine were detected at 304 ngL^{-1} , 339 ngL^{-1} , and 865 ngL^{-1} , respectively. In a study, atenolol, sulfadimethoxine, and sulpiride were found in river sample at 52.9, 77.3, and 164 ngL^{-1} , respectively. In Europe and US in range of mgL^{-1} PPCPs were found in river resource (Lim et al., 2017). In another study, drinking water was analysed to find the presence of phthalates in which di (2-ethylhexyl) phthalate, di-n-butyl phthalate, butyl-benzyl phthalate, and diethyl phthalate were quantified in range of $0.02\text{--}0.6 \text{ }\mu\text{g/L}$ (Cizmas et al., 2017). Phthalates are found in natural water bodies most frequently due to their presence in plasticizers, solvents, and additives (Ji et al., 2010). An earlier study has confirmed the presence of carbamazepine, caffeine, and diclofenac as prominent contaminants in Ukraine and France (Lim et al., 2017). Presence of PPCPs including clindamycin, Caffeine, sulfamethazine, paraxanthine, erythromycin, Lincomycin, and clarithromycin Yangtze river, China were mostly found (Wu et al., 2014). High level ($96.65 \text{ }\mu\text{g L}^{-1}$) of organic compounds such as trimethoprim, paracetamol, diclofenac, caffeine, salicylic acid carbamazepine, and sulfamethoxazole were the dominantly found in WWTPs of Greece (Kosma et al., 2014). In Australia, investigation was performed to identify the presence of PPCPs which revealed contaminants triclosan, venlafaxine, atenolol, sotalol, sertraline, propranolol, metoprolol, carbamazepine, chlorpheniramine, fluoxetine, and diphenhydramine in STPs (sewage treatment plants) (Roberts et al., 2016). Similarly, high concentration of pharmaceuticals was investigated in Indian water bodies (WWTPs, hospital effluents, groundwater, and rivers). The research presented serious results as antibiotic concentration

(sulfamethoxazole, ofloxacin, ciprofloxacin, norfloxacin, and amoxicillin) in WWTPs were higher than present in Australia, North America, Europe, and other Asian countries. Moreover, atenolol, ibuprofen, triclocarban, carbamazepine, trimethoprim, acetaminophen, and caffeine were also identified at high concentration (Balakrishna et al., 2017). A report detected acetaminophen, ciprofloxacin, and caffeine in Pearl River Delta, China at a high concentration of 339, 304, and 865 ngL^{-1} (Fu et al., 2019).

2.5. Environmental factors responsible for resistance

Emerging concern of antibiotic resistance in bacterial strains is not only restricted to persistence of PPCPs but there are other major factors are also considered and identified as potent vehicle for conferring resistance. Global problem related to emergence of ARGs and ARB is getting serious due to the presence of other trace elements which might contribute to even more dissemination of antibiotic resistance. Implementation of products containing estrogenic endocrine disrupting compounds (e-EDCs), toxic metals and nanoparticles in pharmaceuticals and household materials and their subsequent discharge in rivers, pose serious risk to human and veterinary health. Emergence of antibiotic resistance and presence of toxic metals are correlated and can be mediated by transmissible plasmids (Pathak and Gopal, 2005).

For instance, various e-EDCs including 17β -estradiol, nonylphenoldiethoxylate, octylphenol, bisphenol A, octylphenoltriethoxylate, estriolnonylphenol, atrazine, octylphenol, and Nonylphenoltriethoxylate were majorly detected in drinking water, ground water, sediments, waste water and river water samples. Endocrine Disrupting Compounds can be of synthetic or natural origin. e-EDCs are identified as subclass of EDCs which causes the induction of estrogen-like response in human beings and animals (Le Thi Minh et al., 2016). The prolonged exposure of such compounds is correlated with the increased cases of testicular germ line cancer, and breast cancer even at low concentration. Also, intake of EDCs causes cryptorchidism, hypospadias, and decreased sperm count (Vega-Morales et al., 2013).

In aquaculture, products containing metal ions such as copper-containing materials are used as feeding, treating, anti-fouling agents,

and for controlling parasites contamination. Use of these materials in formation of nets and cages and for this purpose copper-alloys, and in commercial fish feeding materials containing lead, mercury, cadmium, and iron are also reported. In addition, other anthropogenic sources also raise alarm in context of aquatic pollution for example toxic elements (cadmium) is found to be present in pesticides and fertilizers leading to direct health related toxicity (Watts et al., 2017). According to a research, sub-inhibitory concentration of toxic metals such as Cu(II), Ag(I), Cr(VI), and Zn(II) are linked to promote conjugative transfer of ARGs in *E. coli* (Zhang et al., 2018a). Evaluation of resistance mechanism employed by toxic elements revealed the incidence of higher mutation rates (Li et al., 2019). Moreover, extensive use of silver nanoparticles (AgNPs) due to their antimicrobial property is also related with environmental toxicity. A study demonstrated the enhanced antibiotic resistance mediated by AgNPs by elevating intracellular reactive oxygen species (ROS) hence inducing stress tolerance (Kaweeterawat et al., 2017).

Therefore, a direct link has been established between antibiotic resistance emergence and antibiotic use in various applications. Increased tolerance of antibiotic has been identified in *Campylobacter* spp., *Salmonella* Spp., *Staphylococcus* spp., and ESBL-producing gram negative bacteria such as *E. coli*.

3. Impact of PPCPs in causing antibiotic resistance to microbes

Microbial evolution occurs due to changing environment and ability to adapt that is essential for bacterial genetic variability to take place. In a way, presence of pharmaceuticals and PPCPs presents selective pressure, enhancing bacterial capability to resist them. Various factors along with role of organic pollutants in exerting resistance are briefly explained in this section.

3.1. Plasmids

Plasmids are involved in horizontal gene transfer (specifically in conjugation) and perform gene transfer from one bacterium to another. Plasmids are considered as small, dispensable, auxiliary chromosome that is self-replicating (Bello-López et al., 2019). Evolution of new antimicrobial resistance and dissemination of resistance genes are majorly contributed by plasmids among various worrisome strains. Notably, in bacterial families including *Enterococcaceae* and *Enterobacteriaceae* conjugative plasmids are identified as drivers of antibiotic resistance (Millan, 2018). Association of bacteria and plasmid are demonstrated to successfully construct “superbugs”. Such specific constructions of clones are pervasive in nature, having antibiotic resistant plasmids linked to specific microbial lineage. For example, *Klebsiella pneumoniae* sequence type 11 (ST11) or ST405 associated with plasmid pOXA-48 (carrying carbapenemase gene bla_{OXA-48}) and *Escherichia coli* ST131 clade C2/H30Rx and IncFII plasmids having beta-lactamase gene bla_{CTX-M} (Millan, 2018).

Plasmids can be categorised into narrow and broad host range. Narrow host range restricts the transfer of plasmids most primarily to related bacterial strains. While broad host, designates the transfer between highly differentiated microbial species. Plasmid RP1 has been identified as potent broad host range plasmid, able to transfer DNA segment to most gram-negative strains and was first found in *Pseudomonas aeruginosa* (Bennett, 2008). Segregation and fitness cost in comparison with conjugation efficiency are the major component for driving fate of plasmid. Though, elimination of plasmid attributed to high growth burden, mating procedures, conjugation machinery, and slow conjugation (Lopatkin et al., 2017). Various antibiotics utilized for clinical purpose have notably acquired plasmid mediated resistance, in which aminoglycosides, fluoroquinolones, and cephalosporin are found commonly. In a study, a novel *Salmonella typhi* resistant to sulfamethoxazole, ampicillin, chloramphenicol, and trimethoprim as well as cephalosporin and fluoroquinolone was identified harbouring a plasmid

encoding bla_{CTX-M-15} (a beta-lactamase gene present in *E. coli*, CTX stands for cefotaximase) extended-spectrum beta-lactamase (Klemm et al., 2018).

Other than natural selection pressure, presence of PPCPs and food preservatives also causes toxicity and antibiotic resistance. For instance, triclocarban, sodium benzoate, and sodium nitrite are found to induce plasmid mediated horizontal transfer of ARGs and conjugative-transfer was found to increase by 2.17–4.31, 6.79–7.05, and 1.24–2.63 folds (Cen et al., 2020).

3.2. Transposable elements

Transposable elements (TEs) or transposons (Tns) are recognized as mobile elements (jumping genes) that can change their location by moving in same DNA or another DNA molecule. Transposase (Tase) are the enzymes, responsible for movement of TEs (Babakhani, 2018). In bacteria, Tns can stay for a longer period and are major cause of antibiotic resistance in various strains. In addition, TEs are identified as major source of emerging resistance in numerous MDR pathogens including ESBL-carrying *Enterobacteriaceae*, methicillin-resistant *Staphylococcus aureus* (MRSA), and vancomycin-resistant *Enterococcus* (VRE) (Rubio-cosials et al., 2018). Conjugative transposons (CTns, also known as integrative conjugative elements) are capable to move from one strain to another and are highly associated with the risk of imposing resistance in most gram-negative bacteria. A CTn named as Tn1549 found to exhibit vancomycin resistance (vanB) in clinical isolates of VRE (van Hal et al., 2016). On the basis of transmission, Tns are categorised into two classes: class I (RTns) and class II (DNA Tns).

Most commonly found conjugative transposons in gram negative bacteria are Tn2350, Tn1525, Tn903, Tn10, Tn9, Tn5 and gram-positive bacteria has been identified to encompass Tn4003 and Tn4001. According to a study, transposon present on *comM* gene in *Acinetobacter baumannii* demonstrated resistance towards antibiotics and toxic metals (Hamidian and Hall, 2018). Another study performed by (Korona-glowniak et al., 2015) identified presence of conjugative transposons in *Streptococcus pneumoniae*. Strain was found to be resistant of clindamycin, chloramphenicol, erythromycin, and tetracycline and showed the presence of numerous Tns. For example, Tn916 carrying *int1*/*xis*Tn916 and *tet(M)*, Tn2010, Tn3872, Tn6002, and Tn2017. In addition, presence of biocides (triclosan, zinc), toxic metals, and antibiotics (sulfamethoxazole, tetracycline, and erythromycin) in aquatic and terrestrial environment can enhance antibiotic resistance via significant increase in ARGs transferred by transposable elements (Gatica and Cytryn, 2013).

3.3. Integrations

In many studies identification of resistant genes revealed that transposons often shared same backbone for conferring antibiotic resistance but more focused investigation demonstrated the presence of another gene cassettes named as integron. ARGs are found to be present on mobile genetic elements such as transposons, integrons, and plasmid. Though, integrons itself do not exhibit mobility its movement depends upon the association with other mobile elements (plasmids and transposons) (Kaushik et al., 2019). Integrons play a key role in development of antibiotic resistance and HGT.

Integrons are capable of capturing and incorporating ARGs into the bacterial genome. This mechanism of integrons is controlled by integron-integrase gene (*intI*). Insertion and integration of exogenous DNA at recombination site of integron is mediated by *intI*. Excision, insertion, and rearrangement of gene-cassettes (circular DNA containing open reading frame (ORF) and cassette recombination site (*attC*) is attributed to integrons (Ghaly et al., 2020). Generally, integrons are categorised on the basis of amino acid sequence of *intI*; segment containing *intI1* are referred to belong class 1, *intI2* referred as class 2, *intI3* identified as class 3, and *intI4* referred to belong class 4 of integrons.

Table 3

Summary of Class1, class2, and class3 integrons, their detection method, and related cassette array reported in previous studies.

Order	Sampling	Species	Integron	Detection technique	Primer	Amplicon size (bp)	Cassette array	References
1.	Broiler carcasses	<i>Escherichia coli</i>	<i>Int11</i>	RFLP-PCR	F-GTTCGGTCAAGGTTCTGG R-CGTAGAGACGTCGGAATG	890	dfrA17-aadA5	Wu et al. (2015)
2.	Broiler carcasses	<i>Escherichia coli</i>	<i>Int12</i>	RFLP-PCR	F-CAAGCATCTCTAGGCGTA R-AGTAGCATCAGTCCATCC	1056	dfrA1-sat2-aadA1	Wu et al. (2015)
3.	Fresh fish	<i>Aeromonas veronii</i>	<i>Int11</i>	PCR	F-GTTCGGTCAAGGTTCTG R-GCCAACCTTCAGCACATG	1958, 1261, 1264, 1618	aadA5-dfrA17-orfC, aadA5-dfrA17, aadA2-dfrA1, aadA2-oxa2	Sarria-Guzmán et al. (2014)
4.	Chicken, fish, clinical samples	<i>Salmonella</i> spp.	<i>Int11</i>	PCR	F-CAGTGGACATAAGCCTGTTC R-CCCGACGCATAGACTGTA	210–230	–	Fabio Ederson Lopes et al. (2014)
5.	Chicken, fish, clinical samples	<i>Salmonella</i> spp.	<i>Int12</i>	PCR	F-TTGGGAGTATCCATAACCTG R-TTACCTGCACTGGATTAAAGC	400	–	Fabio Ederson Lopes et al. (2014)
6.	Farm animals	<i>Salmonella enteritidis</i>	<i>Int11</i>	PCR	F-GCCACTGCGCGTTTACCACC R-GGCCGAGCAGATCCTGCACG	898	drfA1-aadA1, aadA2,	Zhao et al. (2017)
7.	Patient sample (stool)	<i>Shigella</i> spp.	<i>Int11</i>	PCR	F-GGCATCCAAGCAGCAAG R-AAGCAGACTTGACCTGA	Variable	dfrA17/aadA1, dfrA7	Zamanlou et al. (2018)
8.	Patient sample (stool)	<i>Shigella</i> spp.	<i>Int12</i>	PCR	hep74- CGGGATCCCGACGGCATGCA- CGATTTGTGA Hep51- GATGCCATCGCAAGTACGAG	Variable	dfrA1/sat1/aadA1, dfrA1/sat1	Zamanlou et al. (2018)
9.	Clinical isolates	<i>Acinetobacter baumannii</i>	<i>int11</i>	PCR	–	–	ampC, aacA4-catB8, ISAbal-blaOXA-23-GES-14, aadA2-cmlA6-GES-14-qacF, VIM-25-GES-24-qacF, dfrA5-ISAbal-blaOXA-51-blaOXA-40, aadB-GES-11-IMP-1	Goudarzi et al. (2016)
10.	Clinical isolates	<i>Escherichia coli</i> , <i>Klebsiella pneumoniae</i>	<i>Int11</i>	PCR	F-CAGTGGACATAAGCCTGTTC R-CCCGAGGCATAGACTGTA	160	dfrA17-aadA5, dfrA7, dfr2d, dfrA12-orfF-aadA2	Zeighami et al. (2015)
11.	Clinical isolates	<i>Escherichia coli</i> , <i>Klebsiella pneumoniae</i>	<i>Int12</i>	PCR	F-CACGGATATGCGACAAAAAGGT R-GATGACAACGAGTGACGAAATG	788	–	Zeighami et al. (2015)
12.	Swine (pork sample, Feces, skin)	<i>Escherichia coli</i>	<i>Int11</i>	PCR	F-GGCTTCGTGATGCCTGCTT R-CATTCTGGCCGTGGTTCT	146	aadB-aadA1-cmlA6, dfrA17-aadA5, dfrA12-orfF-aadA2	Fang et al. (2019)
13.	Swine (pork sample, Feces, skin)	<i>Escherichia coli</i>	<i>Int12</i>	PCR	F-CACGGATATGCGACAAAAAGGT R-GTAGCAAACGAGTGACGAAATG	789	dfrA1-sat2-aadA1	Fang et al. (2019)
14.	Swine (pork sample, Feces, skin)	<i>Escherichia coli</i>	<i>Int13</i>	PCR	F-GCCTCCGCGACGACTTTCAG R-ACGGATCTGCCAAACCTGACT	980	–	Fang et al. (2019)
15.	Fresh water fish	<i>A. hydrophila</i>	<i>Int11</i>	PCR	F-CAGTGGACATAAGCCTGTTC R-CCCGAGGCATAGACTGTA	160	ant(3'')Ia, aac(6')Ia, dhfr1, oxa2a and/or pse1	Hayati et al. (2018)

Int11: integron class1; *Int12*: integron class2; *Int13*: integron class2; PCR: Polymerase chain reaction; RFLP: Restricted fragment length polymorphism.

Various studies performed to identify integrons in samples and their respective cassette arrays have been reported in Table 3.

The presence of hormones and personal care products also accelerates the resistance mechanism influence by introns. Triclocarban, fluoxetine, ibuprofen, triclosan, caffeine, gemfibrozil, esterone-3-sulfate acetaminophen, 17 β -estradiol, carbamazepine, and naproxen even at low concentration (<5 ng/L) were found to develop antibiotic resistance (Beattie et al., 2018). An investigation was performed to identify class 1 integrase *int11* and was found to contain aminoglycoside resistant genes followed by β -lactams and trimethoprim resistant genes (Ma et al., 2017). Another study demonstrated the role of class 1 integron in *Acinetobacter baumannii*, a nosocomial pathogen. The highest resistance was found against aztreonam, ciprofloxacin, and ceftazidime (Ghazalibina et al., 2019).

3.4. Efflux pump

Acquisition of antibiotics resistance is governed by various mechanisms such as MGEs in which every gene encodes different resistance (co-resistance). Though, few of the determinants are able to confer resistance to different antibiotics (cross-resistance). For instance, efflux pumps are the most conspicuous mechanism to confer cross-resistance. The presence of efflux pumps have been observed in bacterial strains, human cells, and eukaryotic organisms (*Candida albicans*) (Blanco et al., 2016).

Asymmetric accumulation and over-expression of genes are liable to increase efflux (intrinsic) activity boosting the appearance of resistant phenotypes. On the basis of stress response created by toxins or antibiotics, efflux can be efficient and fastest mechanism to combat antimicrobial effect by pumping out drugs out of microbial interior (Du et al., 2018). Six major families contributing to efflux pathways have been identified including major facilitator superfamily (MFS), multidrug and

toxin extrusion (MATE) family, small multidrug resistance (SMR) family, resistance-nodulation-cell division (RND) superfamily, proteobacterial antimicrobial compound efflux (PACE) family and ATP-binding cassette (ABC) family (Santajit and Indrawattana, 2016). In which ABC family works by utilizing ATP as source of energy to perform transportation while other five families are identified as secondary active transporters that works by utilizing electrochemical energy found in ion gradients of transmembrane (Hassan et al., 2015).

(Molale and Bezuidenhout, 2016) performed a study to evaluate the presence of efflux genes (*mefA*, *tetK*, *tetL*, *msrC*) in *Enterococcus* spp. In which 17% presence of *tetL* and *msrC* were identified. Till date, various efflux pumps in pathogenic bacteria have been identified such as *PmrA* in *Streptococcus pneumoniae*, *NorA* in *Staphylococcus aureus*, *MexAB-OprN* in *P. aeruginosa*, and *AcrAB-TolC* in *E.coli*. MDR1, FLU1, CDR1, and

CDR2 genes encoding efflux pump were found to influenced by the presence PPCPs in *Candida albicans* isolates in water sample (Monapathi et al., 2017). Furthermore, interference of PPCPs with efflux pump was identified on zebrafish (*Danio rerio*) embryos. Fluoxetine, nerol, α -hexylcinnamaldehyde, simvastatin, musk xylene, α -amylcinnamaldehyde, isoeugenol, and diclofenac were found to inhibit multi-xenobiotic resistance (MXR) system of fish and also modulated the activity of ABC efflux transporter (Cunha et al., 2017).

3.5. Horizontal gene transfer

Horizontal gene transfer provides a potent route for microbial consortium to gain resistance towards antibiotics and also causes global dissemination of ARGs. The process enabling the transfer of antibiotics

Table 4
Resistance factors and their probable outcomes conferred by resistant pathogenic strains.

Order	Microorganism	Source	Mechanism of antibiotic resistance	Action employed	Gene location/transfer	Antibiotic resistance genes	Antimicrobial resistance Profile	References
1.	<i>Escherichia coli</i>	Clinical isolates	Enzymatic degradation	ESBL production	Chromosomal DNA, plasmid, MGEs	<i>bla_{TEM}</i> , <i>bla_{SHV}</i>	Ampicillin, amoxicillin, trimethoprim/sulfamethoxazole	Anago et al. (2015)
2.	<i>Escherichia coli</i>	Boot swabs and cloacal swab (broiler farms)	Enzymatic degradation	ESBL production	Chromosomal DNA/HGT (plasmid-borne), MGEs	<i>bla_{CTX-M-1}</i> , <i>bla_{CTX-M-2}</i> , <i>bla_{CTX-M-9}</i> , <i>bla_{CTX-M-8}</i> , <i>bla_{TEM}</i> , <i>bla_{SHV}</i>	Ampicillin, cepheims, ciprofloxacin, trimethoprim/sulfamethoxazole	Gundran et al. (2019)
3.	<i>Pseudomonas aeruginosa</i>	Clinical isolates	Enzymatic degradation/modification	Beta lactamases, AAC, ANT, APH	Chromosomal DNA, MGEs, plasmid	<i>bla_{TEM}</i> , <i>bla_{SHV}</i> , <i>bla_{CTX-M}</i> , <i>aadA1</i> , <i>aph(3)-Ib</i> , <i>aph(3)-I</i> , <i>aph(6)-Id</i> , <i>aac(3)-Id</i>	Tobramycin, netilmicin, kanamycin, amikacin, gentamicin, isepamicin, ampicillin, piperacillin, cephalosporins	Subedi et al. (2018)
4.	Multiple strains	Surface water samples	Genetic mechanism and target protection	Efflux pump and ribosomal protection	Chromosomal DNA, plasmid, MGEs	<i>tetA</i> , <i>tetB</i> , <i>tetE</i> , <i>tetM</i> , <i>tetZ</i> and <i>tetW</i>	Tetracycline	Xu et al. (2015)
5.	Multiple strains	WWTPs, hospital and residential area sample	Genetic mechanism	Efflux pump, integrons	Chromosomal DNA, MGEs, non-conjugative plasmid	<i>tetA</i> , <i>tetB</i> , <i>tetC</i> , <i>tetG</i> , <i>tetL</i> , <i>tetM</i> , <i>tetO</i> , <i>tetQ</i> , <i>tetW</i> , <i>tetX</i> , <i>sul1</i> , <i>sul2</i>	Tetracycline, sulfonamides	Li et al. (2015)
6.	Multiple strains	Hospital, WWTP effluent	Target protection and genetic mechanism	Ribosomal protection, efflux pump	Plasmid, MGEs	<i>bla_{TEM}</i> , <i>qnrS</i> , <i>ermB</i> , <i>sul1</i> , <i>tetW</i>	Ciprofloxacin, ofloxacin, cefazolin, cefotaxime, clarithromycin, sulfamethoxazole, azithromycin	Rodriguez-Mozaz et al. (2015)
7.	Multiple strains	Coastal aquaculture	Genetic and enzymatic mechanism	Efflux pump, enzymatic modification	Plasmid, MGEs	<i>tetA</i> , <i>tetB</i> , <i>tetD</i> , <i>tetE</i> , <i>tetG</i> , <i>tetH</i> , <i>tetM</i> , <i>tetQ</i> , <i>tetX</i> , <i>tetZ</i> , <i>tetBP</i> , <i>qnrD</i> , <i>qnrS</i> , <i>aac(6)-Ib-cr</i> , <i>bla_{TEM}</i> , <i>bla_{CTX}</i> , <i>bla_{SHV}</i> , <i>ermC</i> , <i>floR</i>	Tetracycline, sulfonamide, quinolone, β -lactam, macrolides, florfenicol	Jang et al. (2018)
8.	<i>Staphylococcus aureus</i>	Surface water	Genetic and enzymatic mechanism	β -lactamases, Enzymatic inactivation, efflux pump, enzymatic modification	Plasmid, MGEs	<i>bla_Z</i> , <i>van</i> , <i>tetK</i> , <i>tetA</i> , <i>aacA-aphD</i>	Penicillin, vancomycin, tetracycline, aminoglycoside	Foster (2017)
9.	Multiple strains	Clinical isolates, surface water	Genetic and target modification	Altered target, drug efflux, target modification by mutation	Plasmid, MGEs	<i>dfrA</i> , <i>fusA</i> , 32S rRNA gene, <i>gyrA</i> , <i>gyrB</i> , <i>gyrC</i> , <i>rpoB</i>	Dihydrofolate	Melnik et al. (2015)
10.	<i>B.pseudomallei</i> , <i>B. thailandensis</i>	Clinical strains	Enzymatic modification, genetic mechanism	β -lactamases, efflux pump	MGEs	<i>penA</i> , <i>BpeAB-OprB</i> , <i>AmrAB-OprA</i> , <i>BpeEF-OprC</i>	Penicillin, tetracycline, aminoglycosides, marcolides, chloramphenicol, fluoroquinolones, sulfamethoxazole, trimethoprim	Rhodes and Schweizer (2016)

ESBL: extended spectrum β -lactamases; DNA: deoxyribonucleic acid; MGEs: mobile genetic elements; HGT: horizontal gene transfer; AAC: Aminoglycoside acetyltransferase; ANT: Aminoglycoside nucleotidyltransferases; APH: Aminoglycoside phosphoryl transferase; WWTPs: wastewater treatment plants; *B. pseudomallei*: *Burkholderia pseudomallei*.

resistant genes is termed as horizontal gene transfer (McInnes et al., 2020). For example, due to conjugation (transfer of genetic material), HGT was found to be responsible for transfer of plasmid-borne beta-lactamases between microbial strains (Wintersdorff et al., 2016). HGT more specifically occur in area containing high bacterial load though it can occur in any environment such as wastewater treatment plants, soils, human and animal gut (Thanner et al., 2016).

Generally, process of horizontal gene transfer is classified into three major categories including conjugation, transformation, and transduction depending upon the route and mechanism involved. The transfer of ARGs via plasmid (conjugation) has led the resistance of numerous antibiotics such as β -lactams, tetracyclines, quinolones, aminoglycosides, sulfonamides and recently discovered colistin resistance. In case of transformation, various experiments have elucidated the presence of transformed antibiotics (kanamycin, streptomycin, fluoroquinolones, nalidixic acid, and erythromycin) into *Gallibacterium anatis*, *S. pneumoniae*, *Neisseria species*, and *Bacillus* spp. (Wintersdorff et al., 2016). Similarly, dissemination of ARGs has been observed via transduction in *E. coli*, *Salmonella*, *Enterococci*, and *S. pyogenes* having the resistance against gentamicin, tetracycline, kanamycin, β -lactams, and erythromycin (Shousha et al., 2015). Moreover, it has been demonstrated that presence of PPCPs and antimicrobial agents (AMA) provides the acquisition of antibiotic resistance in various microbes (Bello-López et al., 2019).

4. Microbial acquisition of resistance and influence of organic pollutants

Bacteria are most likely to confer resistance via enzymatic machinery and modifying their binding specificity with the target organisms. Capability of microbial strains to synthesize and release enzymes provides them a liability to cope environmental stress. Till date various antibiotics along with their mode of action has been analysed and reported where each of these mechanisms perform different mode of action to resist antibiotic Table 4. Generally, these approaches include:

4.1. Inactivation of antibiotics via enzyme degradation or modification

Various enzymatic mechanisms such as beta-lactams, aminoglycoside resistance-modifying enzymes, Chloramphenicol acetyltransferase, Macrolide-Lincosamide-Streptomycin inactivating enzyme, and tetracycline inactivation have been observed in most antibiotic resistant bacteria.

Primarily activity of β -lactamases is responsible for occurrence of resistance to β -lactam by breaking the amide bond present in β -lactam ring. Many of β -lactam resistant genes are found on integrons, plasmids, and transposons and confer intrinsic resistance. More specifically, β -lactam resistant gene (*bla*) are found on integrons and is the major reason of dissemination of β -lactam resistance (Schillaci et al., 2017). Moreover, extended spectrum β -lactamases has attracted the attention of scientist fraternity due to evolved resistance of third generation cephalosporins. Though initially found susceptible, soon acquired resistance and emerged as global threat due to the spread of ESBL.

O-phosphorylation, O-nucleotidylation, and N-acetylation are the three major mechanisms exhibited by aminoglycoside-modifying enzymes. Modification by aminoglycoside enzymes of antibiotics occurs while movement across cytoplasmic membrane (Lin et al., 2015). Gentamicin, tobramycin, amikacin, and netilmicin are modified by acetyltransferases; tobramycin activity is greatly altered by nucleotidyltransferases; and susceptibility of amikacin is affected by phosphotransferases.

A bacteriostatic inhibitor identified as chloramphenicol acetyltransferase, is the most common modifying enzyme present in gram-positive and gram-negative bacteria for inactivation of chloramphenicol, encoded by plasmid or chromosomal DNA (Kapoor et al., 2017). This enzyme works by utilizing 3-O-acetylation and is essentially

intracellular.

Although MLS (macrolides-lincosamides-streptogramins) resistance is mainly attributed to efflux and ribosomal site alteration, many inactivation enzymes are also been discovered causing resistance to MLS antibiotics. Esterases, phosphotransferases, acetyltransferases, hydrolases, nucleotidyltransferases, and glycosylases encoding genes are found. *E. coli* is the major source of erythromycin esterases, hydrolyses lactone ring and hence inactivation of respective antibiotic. Erythromycin esterases mediated resistance is plasmid encoded and causes high level resistance to erythromycin. Moreover, in *S. aureus* and *Streptococcus haemolyticus*, plasmid-encoded resistance causing adenylation of lincosamides or hydrolysis of streptogramins. According to a study, in *Riemerella anatipestifer* *ereD* and *ErmF* are found associated with erythromycin resistance (Xing et al., 2015).

A flavin-dependent monooxygenase, termed as *tetX*, has been originally identified as only tetracycline-inactivating enzyme. This enzyme was described in *Bacteroides fragilis*, which works on older as well as newer tetracyclines including chlortetracycline, oxytetracycline, minocycline and doxycycline. Determinant of *tet X* has not been found in clinical isolates though a similar gene was isolated from *P. aeruginosa* (Linkevicius et al., 2016). Resistance to tetracycline antibiotics is mainly exhibited by ribosomal protection and efflux pump mechanisms.

4.2. Modification by mutation/protection of target

Changing target structure to prevent efficient binding of antibiotics while retaining normal function of targets can confer antimicrobial resistance. Even the small changes made to target structure subsides the antibiotic attachment to it. Single point mutation created in genes encoding target of antibiotics are found to alter binding between antibiotic target and antibiotic in gene encoding antibiotic. Proliferation of strains can occur while carrying point mutation and thus, can be transfer to another strains (Blair et al., 2014). Like alteration in peptidoglycan structure. Mutation in penicillin binding proteins (PBPs) can alter its binding to β -lactams. Therefore, resistance to ampicillin in *E. faecium*, penicillin resistance in *S. pneumoniae* have been observed (Ogawara, 2015).

Protection of targets by modification demonstrated an effective mechanism for conferring resistance to antibiotics. Although genetic determinants encoding antimicrobial resistance are found in chromosomal DNA of bacteria, most of pertinent genes facilitating resistance are carried by MGEs. Fluoroquinolones (*qnr*), fusidic acid (*fusB*, *fusC*), and tetracycline (*tetM*, *tetO*) are some identified drugs affected by target modification for protection. Investigation of bacterial genome revealed the presence of clinically relevant mechanisms for various relevant antibiotics including erythromycin ribosome methylase (*erm*) family that causes alteration in drug binding site by methylation of 16S rRNA, hence inhibit attachment of streptogramins, lincosamines, and macrolides (Lee et al., 2017).

Besides preventing entry of antibiotics into cell or altering their targets, bacteria can resist antibiotics by destroying or modifying them. Antibiotics can be inactivated by hydrolysis (enzyme catalysis) or by transferring chemical groups. Enzyme-catalysed reactions to breakdown antibiotics are the main mechanism performed by antibiotic resistant bacteria. Numerous enzymes including β -lactams, macrolides, phenicols, and aminoglycosides have been identified for degradation and modification of various classes of antibiotics. β -lactams antibiotics such as penicillin, carbapenems, cephalosporins, monobactams, and clavams can be hydrolysed by diverse variety of β -lactamases. Till date, more than 1000 β -lactamases have been explored. Moreover, different catalytic enzymes including penicillinases, ESBLs (CTX-M), and carbapenemases such as *Klebsiella pneumoniae* carbapenemases (KPC) have been explored so far (Bush, 2015). β -lactamases encoding enzymes are generally termed as *bla* (such as *bla_{KPC}*) and are found in MGEs or bacterial chromosome (Munita and Arias, 2016).

However, various unique properties of ARB contribute to their

growth and persistence in environment. Native bacteria (autochthonous) have reservoir of ARGs known as “resistomes” that can provide resistance to other microbes by subsequent transfer via HGT. Some chemical stressors including toxic metals, PPCPs in aquatic environment are also found problematic as they can induce mutations and selective resistance to microbes. This selective pressure can activate co-selection via co-resistance, cross-resistance, and co-regulation. These stressors can be transcriptionally or genetically linked to other compounds and hence can provide resistance to other compounds (Vikesland et al., 2019).

5. Risk assessment

Discharge of human-used PPCPs in water bodies causes pollution to terrestrial system when employed for irrigation or fertilizer to agricultural land. Veterinary drugs released into environment by exercised for treatment of pasture animal, in aquaculture, manure application from farm facilities. Currently more than 4000 pharmaceuticals are in use for different purposes. Hence, the hazards imposed by PPCPs are uncountable. Although, the regulation and environmental risk assessment strategy depends upon considering every possible source and fate of PPCPs in ecosystem. Dissipation of PPCPs to other natural sources occurs mainly due to formation of non-extractable residues (NERs) (Boxall et al., 2012). NERs are the major reason of increased bioavailability, bioaccumulation, and biomagnification of PPCPs in animals, plants, aquatic mammals, soils, and human. In fact, the potential of PPCPs to magnify and bioaccumulate has been monitored in top organisms of food chain such as carnivorous fish, top predators, large mammals, and birds of prey. According to a report, most vulnerable species to PCPPs found in Arctic including polar foxes, polar bears, arctic birds (glaucous gull), arctic charr, whales, and orcas (Garric et al., 2013).

5.1. Persistence and ecotoxicological impact of pharmaceutical and personal care products on environment

The physiological properties of PCPPs make them persistent and non-degradable by conventional treatment strategies. The inability of treatment processes to remove synthetic organic compounds (SOCs) imposes potential risk to environment, aquatic animals, and human health. However, not all PCPPs are persistent in nature, though the continuous use and disposal into water bodies makes them “pseudo-persistent”, considered as more potent to reside than other organic pollutants. A study (Ebele et al., 2017) has categorised pharmaceuticals and their metabolites as high, moderate, and low persisting compounds in sediments and water according to dissipation time (DT50). In which carbamazepine, clofibrilic acid, and diazepam were found as highly persistent (DT50 $\frac{1}{4}$ 119–328 days), ivermectin, iopromide, and oxazepam were identified as moderately persistent (DT50 $\frac{1}{4}$ 15–54 days) while CBZ-diol, 2-hydroxyibuprofen, paracetamol, and ibuprofen were deemed as low persistent (DT50 $\frac{1}{4}$ 3.1–7 days). Another study classified Oxazepam (anxiolytic drug) as highly persistent in lake water due to immoderate utilization (Klaminder et al., 2015).

Application of personal care products has led its pervasive existence in ecosystem and caused adulteration of water bodies. Ability of poultry farm's antibiotics to dissolve in water is major cause of drinking water contamination. Numerous antibiotics such as chloramphenicol, quinolones, macrolides, ciprofloxacin and sulfonamides have been majorly observed in drinking water with having highest concentration of ciprofloxacin at concentration of 679.7 ng/L (Mceachran et al., 2015). Antibiotics addition to soil can occur via dumping of waste effluents, irrigation, manure application, and landfills. Due to this presence of quinolones, sulfonamides, tetracyclines, and chlorotetracycline have been estimated in detectable concentration ranging from $\mu\text{g kg}^{-1}$ to mg kg^{-1} (Ben et al., 2019). Similarly, in air and dust antibiotics including tetracyclines, tylosin, chloramphenicol, and sulfamethazine have been detected from non-detectable range to 12.5 mg kg^{-1} . Also, monensin, tetracycline, and tylosin were detected from 0.5 to 4.6 mg kg^{-1} in dust

samples (Mceachran et al., 2015).

Moreover, the employment of organic substances in personal care and cosmetics products has caused serious harms. For instance, parabens, a preservative found in PCPs, pharmaceuticals, and food, is designated as endocrine disrupting material (Błędzka et al., 2014). The presence of parabens in aquatic system and hence in aquatic animals has raised alarms. Propylparaben, ethylparaben, methylparaben, butylparaben, isopropylparaben, benzylparaben, and isobutylparaben have been mainly distinguished. Higher concentration of methylparaben (1062 ng L^{-1}), propylparaben (3142 ng L^{-1}) in Chinese river (Błędzka et al., 2014), methylparaben (400 ng L^{-1}) (Kasprzyk-Hordern et al., 2008), propylparaben (69 ng L^{-1}) (González-Mariño et al., 2009) in European rivers, butylparaben (163 ng L^{-1}), and methylparaben (15 ng L^{-1}) in tap water (Casas Ferreira et al., 2011) have been identified. The bioaccumulation study of parabens demonstrated the presence of 4-hydroxybenzoic acid (4-HB) in tissues of marine mammals including liver of dolphins (865 ng/g), and methylparaben and 4-HB in polar bears (Xue et al., 2015).

Although, PCPPs are found in low amount in aquatic environment but being biologically active impose health hazard. Bioaccumulation of such compounds has been identified in various non-target species. For instance, carbamazepine (antiepileptic) has been monitored in *Thamnocephalus platyurus* (crustacean) and *Pseudokirchneriella subcapitata* (algae) with having 12.6 and 2.2 accumulation factor, respectively. Furthermore, 28, 16, 1.4, 16, and 2.0 accumulation factors were measure for ibuprofen, diphenhydramine, carbamazepine, diltrazem, and caffeine, respectively. Similarly, bioaccumulation of antibiotics (trimethoprim, sulfamethoxazole, ofloxacin, sulfamethazine) have been detected in muscle tissues, liver, plasma, and bile of wild fish in Perl river, China (Zhao et al., 2015). Data available on bioaccumulation and biomagnification PPCPs in food webs, divulged the levels and trophic transfer of antibiotics (fluroquinolone, sulfonamide, macrolide, and trimethoprim). Higher bioaccumulation of sulfonamide was detected in benthic fish than pelagic fish and invertebrates, while high level of fluoroquinolones was found in invertebrates. Although, trimethoprim and sulfonamide showed biomagnification in food web, while biodegradation of fluoroquinolone and macrolide was identified (Liu et al., 2017).

Till date very few information reported on effect of PCPPs on birds and small mammals, reptiles, and amphibians. The indirect effect of PCPPs has been analysed on top predators, for example antiparasitic drugs may be associated with adverse effect on insect-eating birds and bats (Boxall et al., 2012). Effect of NSAIDs employed in veterinary sector has been identified to affect Asian vulture population. In addition, ingestion and accumulation of diclofenac, present in animal carcasses has caused birds' death through renal failure (Garric et al., 2013). Risk assessment studies on birds feeding on invertebrates (exposed with STP effluent) validate the presence of EDCs in invertebrates. Depressed immunocompetence, reduced development, decreased growth, and altered immune function has been identified in experimentally exposed adult male starlings (*Sturnus vulgaris*) (Shore et al., 2014). Assuming oral ingestion as only possible route of exposure to PCPPs and EDCs to birds and mammals, three bioaccumulation food web strategies are used 1) from earthworm to eating birds, 2) from fishes to piscivores 3) biomagnification in food web.

Based on environmental pollutants, it is believed that presence of ARGs and PCPPs are associated with chronic toxicity. The accelerated growth and increased level of antibiotics and ARGs due to antibiotic abuse need specific concern. Mainly, risk assessment depends upon acute toxicity data and chronic toxicity levels of organic compounds in environment. Hence, human health and environmental risk assessment need to be thoroughly analysed.

Metabolism of PCPPs has been monitored in plant uptake and drinking water studies. Wu et al. (2010), 2012 demonstrated the presence of carbamazepine (anticonvulsant) in leaves of tomato and soybean but not in tomato fruit and bean of soybean. A report provided by

Goldstein et al. (2014) has proved that metabolites can be detected in fruit. Metabolites (methyl-triclosan) derived from epoxy-carbamazepine and triclosan are found to impose more toxicity than its parent compound itself. Therefore, monitoring of PPCPs metabolism warrants the identification of potential risk caused by such compounds on animals, crops, and environment. Also, existence of PPCPs in ecosystem is associated with increasing temperature that can alter metabolism of ectothermic organism hence altering oxygen demand. Caffeine, a well-known compound to enhance metabolic rate, is associated with altered oxygen demand requirements (Meyer et al., 2019).

The major concern about exposure and persistence of PPCPs are that they pose serious harms to ecology and cause the emergence of ARGs. These ARGs are ubiquitously found in environment and have been found to impart antibiotic resistance to various classes of antibiotics. Presence of ARGs have been monitored in aquatic environment, hospital and industry effluents, livestock wastes, municipal wastewater, drinking water, and surface water (Liu and Wong, 2013b). In addition, multi-antibiotic resistance (MAR) superintegrons have been reported in *E. coli* to confer resistance to sulfonamides, ampicillin, and tetracycline. Concentration of such contaminants was monitored to be high in sediments than in water samples, demonstrating sediments acting as suitable reservoirs for ARGs in ground water (Luo et al., 2010). Furthermore, water-borne PPCPs (diclofenac, 4-tert-nonylphenol, estradiol) at levels of 5–50 µg/L can cause ecotoxicity to aquatic organisms, affecting their kidney, gill, and other immune parameters (Ebele et al., 2017).

5.2. Impact on human health

Pharmaceuticals are water soluble in nature and hence cannot be dissociate in aqueous solution to a significant amount. Pharmaceuticals are basically design to perform physiological and biochemical processes and this property allows them to cross biological barriers and thus making them persistent in human body. Properties of pharmaceutically-active compound to accumulate and induce hazardous effect in non-target species raise serious concerns. Such compounds are associated with adverse pharmacological effect on exposed species. A research conducted to analyze ecotoxic effect of drug orthologs (human) on 16 different species revealed that green alga, daphnids, and zebra fish possessed orthologs to 35%, 61%, and 86% (Roberts et al., 2016).

In addition, use of antibiotic as growth promoters, prophylactic and therapeutic agents in food (egg, milk, meat, vegetables, fruits, and fish) poses health issues. For instance, oxytetracycline, sarafloxacin, thiamphenicol, and erythromycin have been identified in aquatic products. In addition, enrofloxacin (2.0–32.3 µg/kg), ciprofloxacin (2.5–27.5 µg/kg) were detected in vegetables (Li et al., 2014) also distribution of antibiotics in plants have been found in order of root < stem < leaf.

To assess risk on human health, dose of the no observed adverse effect level (NOAEL), (amount of a substance (pharmaceuticals) determined via experiments which causes no detectable adverse impact on target organism) is determined to calculate level of accepted daily intake (ADI). Calculation of ADIs based upon minimum therapeutic dose (MDA) and minimum concentration inducing desirable therapeutic impact on targeted group of people (Adis, 2015). Estimated ADI level can be used to establish a safe and secure human exposure to drugs. Production estimate, toxicity threshold, and production estimate are the three major quantitative risk-based approach to assess human health risk of PPCPs. Adverse outcome pathway (AOP) strategy works on assessing toxicity at each level in order to identify impact of toxic compounds on an organism followed by a whole population (Caldwell et al., 2014). Moreover, investigation on absorption, distribution, metabolism, and excretion (ADME) properties of drugs on human can facilitate a better understanding of dosage and toxicity imposed by respective drug class (Cizmas et al., 2015).

Effect of higher toxicity causing compounds has been demonstrated on endocrine system, progesterone, and testosterone. Various compounds have shown genotoxic properties including lamotrigine,

ambrettolide, sulfamethoxazole, chloramphenicol, 10,11-epoxycarbamazepine, triamterene, sulfapyridine, sulfamethazine, virginiamycin, streptomycin, ciprofloxacin, tylosin, chlortetracycline, tetracycline, norfloxacin, meprobamate, and progesterone (Prosser and Sibley, 2015). Moreover, emergence of ARGs also causes serious issues in line of treatment of fatal infectious diseases. An evidence proves the proliferation and development of antimicrobial resistance escalate antibiotic resistance in medicinal and urban areas (Cizmas et al., 2015). Risk assessment of antibiotic resistance via ARGs and PCPPs is the analysis of antibiotic resistance caused due to prevalence, persistence, and over use of PCPPs products in natural environment. Various concerns of antimicrobial resistance via ARGs and ARB have been elucidated (1) intake of antibiotics causes potential alteration in human microbiome (via ingestion, inhalation, dermal contact) (Guo et al., 2004) (2) selection pressure on microbes due to the presence of antibiotics in natural environment creating reservoir of antibiotic resistance.

Furthermore, most organic pollutants such as polybrominated diphenyl ethers (PBDEs), polychlorinated biphenyls (PCBs), and perfluoroalkyl substances (PFASs) are specifically designed to work efficiently in biological system even at low concentration. These compounds work to modify or accelerate cell-signalling, metabolic, and enzymatic pathways. Hence, the occurrence of PCPPs in aquatic environment enhance their potential to interfere with endocrine system, causing disruption of homeostasis and other adverse effect (Ebele et al., 2017).

6. Current dilemma and discussion

Execution of pharmaceutical and antibiotic therapy in medical use has been identified as one of the most fortunate discovery of modern age, providing a crucial system for complicated and sophisticated clinical intervention allowing us to efficiently enhance the life span of people around the world. Therefore, the usage, excretion, and disposal of such persistent organic compounds in aquatic system have raised inadvertent presence in environment. Hence, in an indistinguishable form for survival, bacteria have also achieved evolution by developing complex mechanism in order to circumvent the attack of antimicrobial agents.

Studies have shown the sources like pharmaceutical industries, municipal wastes, house-hold, veterinary, aquaculture, anti-fouling paints and agriculture related effluents has caused a big influence on emergence of various ARB and ARGs. Even the low concentration, these compounds are capable of inducing maximum biological activity for a prolonged period. This review highlights the presence of pharmaceuticals, endocrine disruptors, and personal care products in soil, sediments, aquatic system, and humans throughout the life-cycle and mounting detrimental effects. Despite several advanced degradation techniques, metabolites of these hazardous compounds are still found in environmental matrices.

In our study we found that PPCPs demonstrated a significant effect on terrestrial and aquatic environment, altering bacterial communities and other residing species. Prevalence of antibiotics including tetracycline, beta-lactams, erythromycin, ceftazidime, amoxicillin, cefepime, cefpodoxime, aztreonam, cefotaxime, sulphonamides, doxycycline, fluoroquinolones, penicillins, lincomycin, trimethoprim, bacitracin, colistin, chloramphenicol, cephalosporin, and many more in waste water and natural water bodies have provided resistance to most microbial communities. Similarly, PPCPs also associated with certain highly deleterious effects. Although, it is apparent from the review that there is a considerable need of more research to assess the effect of PPCPs on humans and other vertebrates. As, the organic pollutants that are less studied have showed dangerous responses. As most toxicological studies have revealed the occurrence of hormones, anti-depressants, anti-inflammatory, and analgesics in water resource. Very few studies have been performed to identify the presence of synthetic husks/fragrances, a common compound of food and personal care products.

Similarly, studies to identify antioxidants are also need to be accelerated. Publication indicates a knowledge gap regarding detailed information about antimicrobial properties and bioaccumulation of PPCPs. This review also demonstrated the need of more researches focused on PPCPs in Indian aquatic resources. As being highly populated country after China, Indian water bodies are identified as major reservoir for organic pollutants, ARGs, and ARB.

This review suggests the implementation of congruent guidelines of antibiotic and PPCPs use in daily basis worldwide. Appearance of antibiotic resistance can be more dangerous and rapid in countries where consumption is higher. Therefore, measures regarding optimal use of antibiotics in treatment of infections and according to age should be considered. In addition, global and domestic policies should be imposed to stop the misuse and overuse of antimicrobial agents. Also, future research should be focused on complete degradation of parent compounds along with their metabolites. For instance, indole, a common fragrance present in most coffees and teas, is found recalcitrant in nature to be degraded and hence bioaccumulation in earthworms have been reported. This is very important as such metabolites produced from natural degradation via UV-light, temperature, pressure can produce more toxic and unknown effect in non-target species as well as in humans.

7. Conclusion

Due to the widespread utility of pharmaceuticals and personal care products in human, veterinary, agriculture, horticulture related aspects as well as laboratories executing genetic modifications has caused evolutionary pressure to enable appearance of resistant strains. To identify apparent cause of resistance developed for clinically important drugs, biotechnologists have illustrated potential strategies conferring antimicrobial resistance. Unregulated employment of PPCPs and related products has posed serious environmental and human health related emergencies. Therefore, risk assessment and subsequent management of organic compounds is the key to subside and counter these pollutants.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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